

An Abstract Form Guided Method for Image Generation without Training

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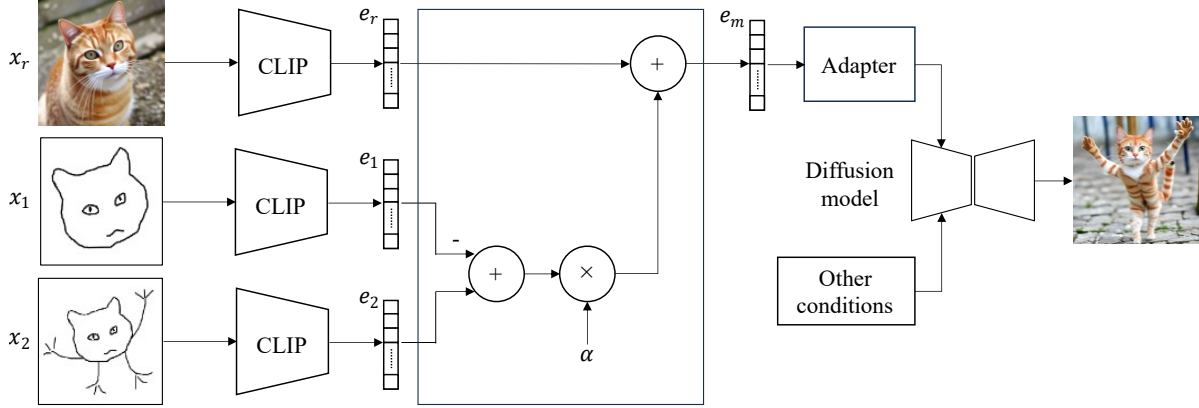


Figure 1: Abstract form guided method for image generation.

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1 Introduction

Image generation techniques play a critical role in various fields. However, existing methods face several challenges. Text prompts, commonly used in diffusion models for controlled image generation [Rombach et al. 2022], often struggle to precisely express certain concepts, such as specific shapes or colors. Moreover, finding suitable text prompt combinations can be time-consuming. ControlNet [Zhang et al. 2023] relies on reference images to control generation but requires additional training to adapt to new control styles. Moreover, generating an exact control condition may be challenging.

To address these limitations, we propose a novel image generation method guided by flexible abstract forms. Our approach enables

users to generate new images based on reference images and modifications to their abstract representation forms, such as scribbles, sketches, or emojis. New images are generated by semantically following the modifications, which enables flexible abstract forms and modifications. Meanwhile, abstract forms guided image generation ensures more precise control than text prompts in terms of shapes or colors. Our approach applies modifications to the latent space of a pre-trained CLIP [Radford et al. 2021] model, and no training is required. IP-Adapter [Ye et al. 2023] is utilized for guided image generation based on modified CLIP representations. This method offers more precise control over image generation compared to text prompts and adaptability compared to ControlNet.

The contributions of this paper are as follows:

- We introduce a training-free method for controlled image generation using abstract forms.
- We demonstrate that our method can flexibly handle various abstract styles, such as scribbles and emojis.
- We demonstrate that our approach can complement existing text- and image-guided generation techniques, thereby enhancing the precision of generated images.

2 Our Approach

Our approach begins with a reference image, from which we extract its abstract form and make modifications. The reference image and its abstract forms are then converted into embeddings in the latent space using a pre-trained CLIP model.

Let x_r be the reference image. Let x_1 be the abstract form of x_r , and x_2 be the image with modifications applied to x_1 . A CLIP

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Figure 2: Comparison of image generation methods.

encoder converts x_r , x_1 , and x_2 into embeddings e_r , e_1 , and e_2 , respectively.

A naive approach would be to directly merge the vectors e_r and e_2 , incorporating all semantic components from x_2 , including image styles and modifications, into x_r :

$$e_m = e_r + \alpha \cdot e_2 \quad (1)$$

Here, α represents the weighting factor. However, the semantic components in x_2 (e.g., style) may conflict with those in x_r , resulting in poor generation quality.

To address this issue, we propose modifying the embedding of the reference image using the difference vector between the embeddings of the abstract images:

$$e_m = e_r + \alpha \cdot (e_2 - e_1) \quad (2)$$

The difference vector $e_2 - e_1$ cancels out semantic components that are not related to the modification, leading to fewer conflicts between $e_2 - e_1$ and e_r . By adjusting the value of α , we can strengthen or weaken the influence of the modification. The modified embedding e_m is then sent to the adapter module in IP-Adapter to guide image generation. The model framework is shown in Figure 1.

3 Experiment

3.1 Comparison with Existing Methods

In the first experiment, we compare our method with existing approaches within the framework of IP-Adapter. For this comparison, we use scribbles or emojis as abstract forms. It is essential to note that in our approach, the abstract forms do not need to precisely match the objects in the reference images, and the modifications do not require pixel-perfect accuracy.

The results are shown in Figure 2. The fourth to sixth columns display images generated using text prompts, ControlNet (with x_2 as the condition), and the naive approach described in Equation 1, respectively. The last two columns show images generated using our proposed method, with the final column incorporating additional text prompts.

The results demonstrate the following:

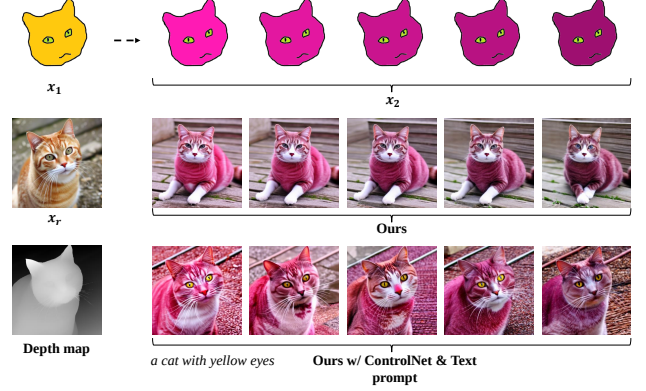


Figure 3: Abstract form-guided image generation for color changes.

- **Text prompts:** While text prompts can guide image generation, they often fail to produce images with specific shapes or colors that match the intended modifications.
- **ControlNet:** ControlNet generates images that strictly adhere to the modified abstract forms, often resulting in irregular or unnatural objects due to pixel-by-pixel alignment.
- **Our method:** Our approach is flexible enough to produce visually reasonable images that follow the modifications without being overly constrained. Furthermore, combining our method with text prompts enhances the quality of the generated images.

3.2 Controlling Color Changes

In the second experiment, we evaluate the ability of our method to control color changes. The results are presented in Figure 3.

The first column in Figure 3 shows the reference image, its abstract form, and the depth map. The depth map is used as a condition for shape control in ControlNet. The modified scribbles (x_2) represent different color changes. The second row contains images generated using our method, while the third row shows images generated by combining our method with ControlNet and specific text prompts.

The results highlight the following:

- Our method accurately reproduces the color changes specified in the abstract forms.
- When combined with ControlNet, our method can simultaneously control both shape and color, producing high-quality results that align with the intended modifications.

4 Conclusion

We proposed a training-free method for controlling image generation using abstract forms like scribbles and sketches. This approach addresses the limitations of text prompts, allowing for precise expression of shape and color.

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